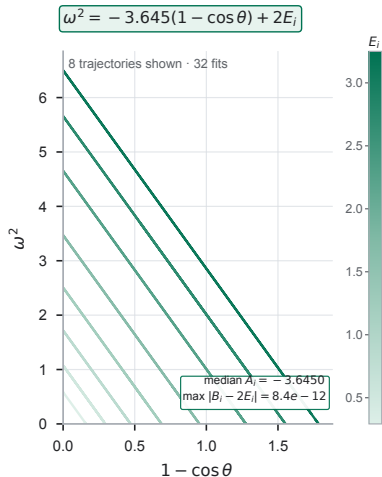


a Task contracts select trajectory representations

32 realizations · compression invariant and protected autonomous prediction

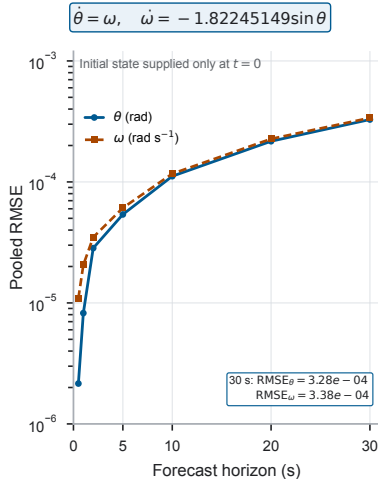
Compression

Shared invariant + one scalar per realization



Prediction

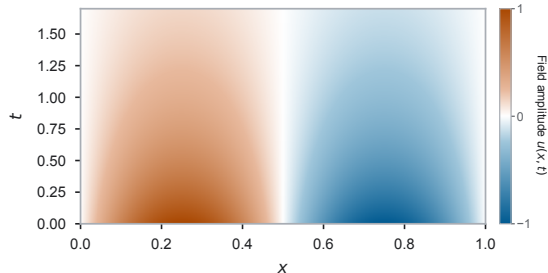
Frozen law + eight protected holdouts



b Structural uncertainty supports multiple representations

Temporal, spatial and coupled relations on one mode

Observed field $u(x, t)$



Temporal representation

$$u_t + \alpha u = 0$$

$$\|R\|_\infty = 1.2e-05$$

$$\alpha = 1.18435$$

Spatial representation

$$u_{xx} + (m\pi)^2 u = 0$$

$$\|R\|_\infty = 0.0e+00$$

$$m = 2$$

Coupled diffusion representation

$$u_t - \kappa u_{xx} = 0$$

$$\|R\|_\infty = 1.2e-05$$

$$\kappa = 0.03$$

All displayed relations: $\|R\|_\infty < 10^{-4}$